The factor objective under the mandate: what a budget of tracking error buys on French's 49 industries

Carlo Hofer. November 2026. The second note of the project Optimal versus naive diversification; the first, of October 2026, covers the replication and the mandate. Code and notebooks: github.com/ochofer/optimal-vs-naive-diversification.

The first note found that under a long-only index-tracking mandate on French's 49 industries the covariance estimator still matters by about a tenth of the tracking error, and that the mandate's industry tilt leaves small factor exposures behind that nobody chose. This note turns the question round: each month from 1979-07 to 2026-08 the optimiser spends a budget of forecast tracking error, 75, 100 or 150 basis points a year, on the factor objective, the sum of its active exposures to value, profitability, investment and momentum in beta units, under every constraint of the mandate, with the forecast from one of five covariance estimates. Seven expectations were written before any path ran; three are met and four are not, each reported with its arithmetic. At 100 basis points the budget binds in 96.6 to 97.3 per cent of the months the mandate alone can meet and buys 0.109 to 0.140 of exposure to the four factors together, 74 to 85 per cent of it in profitability and investment, at a realised tracking error of 118.6 to 137.0 basis points a year on the daily estimators, 19 to 37 per cent above the budget, and 150.1 on the monthly covariance. The targeted factors contributed +26.7 to +35.5 basis points a year, beyond two standard errors, and the industries' own returns took back most of the factor part, so that no path's active return is beyond two standard errors of zero; the six factors explain 14 to 28 per cent of the active return's variance, where 30 was expected, because the factor part and the residual move against each other. The estimator matters in what the budget cost, 18.4 basis points across the four daily estimators, and within 4.5 per cent in what it bought.

Terms used in this note, each with the one meaning it carries throughout.

value-weighted: said of French's industry returns: each firm counts in proportion to its market capitalisation, the number of its shares times their price, at the start of the month. **benchmark**: the index the portfolio tracks, the cap-weighted (weighted by market capitalisation) combination of French's 49 value-weighted industry portfolios. **active weight**: an industry's portfolio weight minus its benchmark weight. **active return**: the portfolio's return minus the benchmark's. **tracking error**: the standard deviation of the monthly active return times the square root of 12; forecast when computed from a covariance matrix before the month, realised when measured afterwards. **basis point**: a hundredth of a percentage point. **excess return**: a return minus the one-month Treasury bill rate. **covariance matrix**: the table of the variances of the assets' returns and the covariances between them. **covariance estimator**: a method that turns a window of returns into a covariance matrix; the five here are the sample covariance, Ledoit-Wolf shrinkage, a six-factor model and a principal-component model on three years of daily returns, and the sample covariance of 120 monthly returns, with the robust portfolio, whose budget holds under all five at once, and RiskMetrics, an exponentially weighted covariance, as sensitivities. **principal component**: a weighted combination of the assets' returns that captures the largest variance not captured by the components before it; the principal-component estimator builds the covariance from the benchmark's return and four of them. **shrinkage**: pulling an estimate towards a simpler target; Ledoit-Wolf shrinkage pulls the sample covariance towards a scaled identity matrix. **mandate**: the whole set of rules the portfolio obeys: long-only, each industry within 2 percentage points of its benchmark weight, one-way turnover of at most 2 per cent a month, Coal, Oil and Utilities at no more than half their benchmark weight (the tilt), tobacco and weapons excluded, and a beta of one to the benchmark. **tilt**: a constraint holding named industries below their benchmark weight, here Coal, Oil and Utilities at no more than half of it. **one-way turnover**: half the sum of absolute weight changes at a rebalance, the fraction of the portfolio sold; the turnover cap is its upper limit. **factor**: a return series that many assets move with; here the market (Mkt-RF), size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum (Mom) factors of Fama and French and Carhart. **beta**: the slope of an asset's return on a factor's return, or on the benchmark's; here estimated on the three years of daily returns before each month. **active exposure**: the portfolio's beta to a factor minus the benchmark's, the sum over industries of the active weight times the industry's beta to the factor, in beta units. **premium**: the average return of a factor over long samples, as the literature reports it; the study estimates none. **targeted factors**: the four of the six factors with a documented premium, value, profitability, investment and momentum, each counted in the direction of its premium and weighted equally in beta units. **factor objective**: the quantity version 2's optimiser maximises: the sum of the portfolio's active exposures to the targeted factors. It is not a tilt; the tilt keeps its one meaning, the industry reduction. **budget**: the limit on the forecast tracking error of the factor objective's portfolio, 75, 100 or 150 basis points a year; a month in which the mandate's tracking-error-minimising portfolio already exceeds it takes that portfolio and is counted. **incidental exposure**: the active exposure the mandate's tracking-error-minimising portfolio of the same month carries with no objective, what the tilt leaves behind. **path**: the month-by-month sequence of portfolios of one covariance estimate at one budget with one set of targeted factors, each built on the data before its month. **main path**: one of the fifteen paths of the five covariance estimators at the three budgets; the robust, RiskMetrics and single-factor paths at 100 basis points are sensitivities. **bias statistic**: the standard deviation of the monthly active return divided by the same month's forecast (the forecast tracking error over the square root of twelve); 1 for a calibrated forecast, above 1 an under-forecast. **standard error**: the standard deviation of an estimate; for a mean active return over 47 years, the realised tracking error over the square root of 47. **attribution**: the split of a month's active return into the six contributions, each the active exposure to the factor times its return in the month, and a residual, the active return minus the six, the part the industries' own returns account for. **targeted total**: the sum of the four targeted factors' contributions, per year. **factor share**: one minus the variance of the residual over the variance of the active return, the share of the active return's variance the six contributions explain. **exposure bought**: a path's mean active exposure per basis point of the tracking error it realised; per factor, the mean

active exposure to that factor over the realised tracking error in basis points a year. **gap**: for a month, the factor objective chosen with the betas estimated before the month minus the value the same active weights carry with the betas estimated after it; a path's gap is the mean absolute value, and the gap per unit divides it by the path's mean factor objective. **quadratic program**: an optimisation whose objective is quadratic and whose constraints are linear, which the solver reaches exactly; a cone program adds a constraint written as a cone, which an interior-point method reaches to a set precision. **expectation**: a statement about the results written and dated before the code that tests it ran; its verdict is MET or NOT MET. **vintage**: the version of French's files on the download date, named by the CRSP cut French built them from; every number here is on the 202608 vintage. **provenance**: the record of the download date, the files' hashes and the vintage that every number was computed on, and, for a run, the solver's name and version.

1. Introduction

An enhanced index fund with a sustainability mandate is a portfolio whose tracking error has a price. The fund holds every industry within a bound of its benchmark weight, trades no more than a cap per month, holds named industries below their benchmark weight and excludes others, and the tracking error these rules produce is the cost its client agrees to bear (Robeco, 2024). The first note of this project (Hofer, 2026), a replication of DeMiguel, Garlappi and Uppal (2009) extended to such a mandate, measured that cost on French's 49 industry portfolios: the tracking-error-minimising portfolio realises 84 to 96 basis points a year, the choice of covariance estimator moves it by about a tenth, the question Dom, Howard, Jansen and Lohre (2024) asked of the minimum-variance portfolio, and the tilt against three industries leaves exposures of about 0.01 to the common factors that nobody chose and nothing in the mandate pays for. The question this note asks is the one a holder of that mandate asks next: what a budget of that tracking error buys when it is spent on purpose, on exposure to the factors with a documented premium, and what the exposure earned.

The existing evidence answers a neighbouring question. The factor literature documents the premia and their long-sample averages (Fama and French, 2015; Carhart, 1997), and the practitioner literature on enhanced indexing describes portfolios that take factor exposures within a tracking-error limit (Robeco, 2024). What neither gives is the arithmetic of the purchase under a given mandate: how much exposure in beta units a basis point of forecast tracking error buys when the active weight bound is 2 percentage points on 49 industries, how far the realised tracking error exceeds a budget that the optimiser's own forecast was held to, which of four equally weighted factors the optimiser buys when the industries' betas to them differ, and how much of the exposure chosen on the betas of the past the betas of the following month take back. Nor does it say what share of the resulting active return the factors explain, against the industries' own returns, when the assets are industries and not stocks.

This note runs the factor objective under the mandate. Each month from 1979-07 to 2026-08 the optimiser maximises the sum of the portfolio's active exposures to value, profitability, investment and momentum, each counted in the direction of its premium and weighted equally in beta units, under every constraint of the mandate and a budget on the forecast tracking error of 75, 100 or 150 basis points a year, with the forecast from one of the five covariance estimates of the first note; the betas are those of the three years of daily returns before the month. Nineteen paths run: the five estimators at the three budgets, the fifteen main paths, and four sensitivities at 100 basis points, the robust portfolio, RiskMetrics, and the sample covariance targeting value alone and momentum alone. Seven expectations were written before any path ran, two of them restated from the arithmetic of a static run at three months, before the rolling run, with the earlier levels kept and marked. The design estimates no premium: the factors' contributions are their realised returns over these 566 months times the exposures the budget bought, and the note reports the active returns with their standard errors and says of none that it is beyond two of them. An attribution then splits each path's active return into the six factors' contributions and a residual, and two measures named before the code, the exposure bought per factor and the gap per unit of exposure bought, say what the budget was spent on and how much of it the next month's betas take back.

French's 49 industries are an informative setting for the question because the first note has already measured the mandate without the objective on the same months, so every figure here has its counterpart: the incidental exposures the tilt leaves behind, the tracking error the mandate costs on its own, the bias of each estimator's forecast and the attribution of the mandate's active return. At the same time, the industries' betas to the four factors differ by about 0.5, so an active weight bound of 2 percentage points allows exposures of 0.03 to 0.08 per factor, large enough to measure month by month and small enough to be what a mandate allows, and the budget is written as a second-order cone on the forecast, so the problem the optimiser solves is a cone program where version 1's was a quadratic program, which changes the arithmetic of a rerun and is recorded with it.

Empirically, the budget bought what it was asked to buy and the purchase was uneven: at 100 basis points it binds in 96.6 to 97.3 per cent of the months the mandate alone can meet and buys 0.109 to 0.140 of exposure to the four factors together, 0.045 to 0.056 of it in profitability and 0.038 to 0.063 in investment against 0.018 to 0.024 in value and 0.002 to 0.009 in momentum, at a realised tracking error of 118.6 to 137.0 basis points a year on the daily estimators and 150.1 on the monthly covariance, 19 to 50 per cent above the budget. The targeted factors contributed +26.7 to +35.5 basis points a year at 100 and +49.9 to +58.9 at 150, beyond two standard errors on all ten of those paths, the industries' own returns took back 31 to 37 of the factor part's +37 to +47 at 100, and no path's active return is beyond two standard errors of zero; the six factors explain 14 to 28 per cent of the active return's variance where 30 was expected, because the factor part and the residual move against each other. What the estimates exclude is a zero targeted total at 100 and 150 basis points, where the sum is 2.1 to 3.4 standard errors from zero, and a difference between the daily estimators in what the budget bought beyond 4.5 per cent of its mean; they do not exclude a zero active return on any path.

2. Design

The data and the mandate are those of the first note: French's 49 value-weighted industry portfolios and his six factors, downloaded at run time on the 202608 vintage, the cap-weighted benchmark rebuilt from French's firm counts and average sizes, and the mandate of long-only weights within 2 percentage points of the benchmark's, one-way turnover of at most 2 per cent a month, Coal, Oil and Utilities at no more than half their benchmark weight, tobacco and weapons excluded, and a beta of one to the benchmark. The sample is July 1979 to August 2026, fixed in the code; the recorded tables are those of the 202608 vintage as cached, and a run on a later vintage, cut at August 2026, is a confirmation run, labelled by its vintage in the provenance record and expected to reproduce every verdict line and every printed decimal and not every digit, because French revises history. Every table this note cites is committed in the repository's `results/202608/` folder with the provenance records of French's files and of the solver.

The factor objective replaces the first note's objective and keeps its constraints. Each month the optimiser finds the weights that maximise s'B'(w minus b), where w are the portfolio weights, b the benchmark's, B the 49 by 6 matrix of the industries' betas to the six factors estimated on the three years of daily excess returns before the month, and s the target vector, 1 on value, profitability, investment and momentum and 0 on the market and size, so that the objective is the sum of the four targeted exposures in beta units; the weights are long-only, within the active weight bound, within the turnover cap from the previous month's drifted weights, under the tilt, the exclusions and beta neutrality to the benchmark from the covariance in use, and under the budget, the forecast tracking error from that covariance at or below 75, 100 or 150 basis points a year. The budget is a second-order cone constraint, so the problem is a cone program, solved by an interior-point method to a set precision where the first note's quadratic program was reached exactly; a month in which the mandate's tracking-error-minimising portfolio already exceeds the budget takes that portfolio and is counted as a month the budget could not be met, and the turnover cap is relaxed to the smallest feasible turnover in the months in which the benchmark itself moves more than the cap allows, as in the first note. The function was tested against an independent solver on known answers before any path ran, and a static run at the three report months of the first note gave the arithmetic on which two expectations were restated.

Nineteen paths run over the 566 months. The five covariance estimates of the first note, the sample covariance, Ledoit-Wolf shrinkage (Ledoit and Wolf, 2004), a six-factor model and a principal-component model on three years of daily returns and the sample covariance of 120 monthly returns, run at each of the three budgets, the fifteen main paths; at 100 basis points the robust portfolio, whose budget holds under all five covariances at once, and RiskMetrics run as sensitivities to the estimator, and the sample covariance targeting value alone and momentum alone as sensitivities to the target. The seven mandate paths of the first note are rebuilt beside them as the comparison. Each path is evaluated on the realised tracking error, the mean forecast, the ratio of the realised tracking error to the budget, the bias statistic, the gross active return with its standard error, one-way turnover and the trading cost at 50 basis points per unit of turnover, the months in which the budget binds (the forecast within one basis point of it), is not met, and in which the turnover cap binds, the mean active exposure to each targeted factor and the exposure bought per basis point of realised tracking error. The gap is recorded month by month.

Seven expectations were written on 8 October 2026 in the constants file, before any path ran, and Table 5 states them with their verdicts. Two were restated on 9 October from the static run's arithmetic, before the rolling run: E1's denominator became the months the mandate alone can meet, because the static run found the mandate's own minimum

forecast above 100 basis points in 22 to 43 of the 566 months by estimator, nearly all in 1979 to 1989, so the budget cannot bind there whatever the objective; and E2's level became 0.024, four fifths of the static run's 0.030, the fifth being an allowance for what the turnover cap, the infeasible months and the rolling betas may cost against the stationary solution. The levels of 8 October stay in the file marked as superseded. The verdicts are strict, MET or NOT MET, and a verdict not met is reported in the same words as one met, with its arithmetic; an explanation of a miss stands beside the verdict and never in its place. Two measures were named in the constants on 8 November 2026 before the attribution's code, with no expectation attached: the exposure bought per factor and the gap per unit.

The attribution rebuilds the nineteen paths with their weights, exactly as the first note rebuilt the mandate's, checks that they reproduce the recorded realised tracking errors within 0.1 basis point and the committed monthly records within 1e-8 per cell, and splits each month's active return into six contributions, the active exposure to each factor with the betas estimated before the month times the factor's return in the month, and a residual, the industries' own returns; the identity, contributions plus residual equal the active return, is checked in every month. The targeted total is the sum of the four targeted contributions, the quantity E4 asks about, and the factor share is one minus the variance of the residual over the variance of the active return, as the first note defined it. Because the share can be read in more than one way, the active return's variance is also split into three parts that sum to one, the factor total's own variance, the residual's own variance and twice their covariance, so that a reader can judge either definition. The recorded run of each notebook is one from a blank folder on the cached 202608 files, twice, byte-identical; the rule for every rerun, written on 6 and 7 November 2026, is that a table from the quadratic programs reproduces to the last digit, the monthly records of a cone program to 1e-8 per cell and its tables of measures to 1e-7, with the solver's name and version in the provenance record and the months the solver reports as inaccurate counted beside the comparison and not in it.

3. Results

3.1 What the budget bought

Table 1 reports the nineteen paths and Table 2 the exposures they bought. At 100 basis points the budget binds within one basis point in 96.98, 97.32, 96.58 and 96.88 per cent of the months the mandate alone can meet, for the sample covariance, shrinkage, the factor model and principal components, against the 95 per cent E1 asked for (met); the mandate alone cannot meet the budget in 37, 43, 39 and 22 of the 566 months, nearly all in 1979 to 1989, when Oil held 15 per cent of the benchmark and the tilt's own forecast exceeded 100 basis points, and the paths miss it in a further 10, 7, 11 and 11 months under the turnover cap. At 75 the mandate alone cannot meet the budget in 100 to 132 months, and at 150 in none for the daily estimators and two for the monthly covariance, and there the mandate's bounds, and not the budget, stop the move in 104 to 201 of the 566 months on the five estimators. The four targeted exposures are positive on every estimator at 100: profitability +0.045 to +0.056, investment +0.038 to +0.063, value +0.018 to +0.024 and momentum +0.002 to +0.009, an average of 0.0289 for the sample covariance against the restated level of 0.024, and the average rises with the budget on every estimator, from 0.012 to 0.020 per factor at 75 to 0.046 to 0.051 at 150 (E2 met). Against the mandate's incidental exposures, minus 0.015 to minus 0.010 in value, minus 0.012 to minus 0.008 in profitability, minus 0.008 to minus 0.005 in investment and minus 0.005 to minus 0.004 in momentum, the objective turned all four round and bought three to five times their size at 100 basis points.

The purchase is uneven. Per basis point of realised tracking error the five estimators buy 0.00092 to 0.00096 of exposure to the four together at 100, and the four daily ones within 4.5 per cent of each other, the second clause of E7; the split is 0.00038 to 0.00040 to profitability, 0.00030 to 0.00042 to investment, 0.00012 to 0.00018 to value and 0.00001 to 0.00007 to momentum, so profitability and investment take 74 to 85 per cent of what the budget bought. The four were weighted equally in the objective; the optimiser bought most of the two whose industry betas it could raise within the mandate's bounds at the least forecast tracking error, and least of momentum, whose industry betas are the smallest and rotate fastest, so an equal weighting in beta units is not an equal purchase. The two single-factor paths show the cost of a single target: value alone buys 0.00060 of value per basis point and gives up 0.00030 of investment, and momentum alone buys 0.00032 of momentum and gives up value and investment, so that its exposure to the four together is 0.00002 per basis point, a twentieth of the four-factor paths'. The monthly covariance's path buys the most investment and the least momentum, 0.00001.

Table 1. The nineteen paths of the factor objective, 1979-07 to 2026-08 (566 months): tracking error, the forecast's miss, active return and trading

Covariance estimate	*Realised tracking error, bp a year*	*Mean forecast, bp a year*	*Realised over budget*	*Bias statistic*	*Active return, gross, bp a year (s.e.)*	*Trading cost at 50 bp, bp a year*	*Active return, net, bp a year*	*One-way turnover, % a month*	*Budget binds, months*	*Turnover cap binds, months*	*Budget not met, months*
Budget of 75 basis points a year											
Sample covariance, 3 years daily	100.2	79.2	1.34	1.22	-2.2 (14.6)	10.7	-12.8	1.78	449	330	116
Ledoit-Wolf shrinkage, 3 years daily	98.6	79.8	1.31	1.18	-3.1 (14.4)	10.3	-13.4	1.72	428	291	137
Six-factor model, 3 years daily	110.2	79.4	1.47	1.34	+3.7 (16.1)	9.0	-5.3	1.49	436	200	129
Principal components, 3 years daily	112.7	78.1	1.50	1.39	-3.9 (16.4)	10.3	-14.1	1.71	457	304	109
Sample covariance, 120 months	128.0	79.6	1.71	1.55	-0.9 (18.6)	11.8	-12.7	1.97	468	470	97
Budget of 100 basis points a year											
Sample covariance, 3 years daily	120.6	101.1	1.21	1.18	+8.9 (17.6)	11.5	-2.6	1.92	513	447	47
Ledoit-Wolf shrinkage, 3 years daily	118.6	101.2	1.19	1.16	+6.5 (17.3)	11.3	-4.8	1.88	509	418	50
Six-factor model, 3 years daily	133.6	100.7	1.34	1.32	+15.5 (19.5)	10.7	+4.8	1.78	509	378	50
Principal components, 3 years daily	137.0	100.8	1.37	1.35	+9.9 (19.9)	11.2	-1.3	1.87	527	427	33
Sample covariance, 120 months	150.1	101.8	1.50	1.45	+6.0 (21.9)	12.0	-6.0	2.00	517	518	47
Robust (worst case of the five)	115.6	102.7	1.16	1.08	+2.3 (16.8)	11.2	-8.8	1.86	493	410	73
RiskMetrics (exponentially weighted), 5 years daily	118.6	101.7	1.19	1.15	+7.5 (17.3)	11.8	-4.3	1.96	512	491	48
Sample covariance, 3 years daily, value alone	122.8	101.0	1.23	1.20	-14.5 (17.9)	11.2	-25.7	1.87	507	408	47
Sample covariance, 3 years daily, momentum alone	123.0	101.0	1.23	1.21	+7.8 (17.9)	11.6	-3.9	1.94	518	478	46
Budget of 150 basis points a year											
Sample covariance, 3 years daily	173.1	145.7	1.15	1.20	+33.9 (25.2)	11.9	+21.9	1.99	410	509	7
Ledoit-Wolf shrinkage, 3 years daily	171.3	146.7	1.14	1.18	+34.0 (24.9)	11.9	+22.1	1.98	430	510	8
Six-factor model, 3 years daily	187.7	143.1	1.25	1.31	+33.9 (27.3)	11.5	+22.3	1.92	356	469	9
Principal components, 3 years daily	185.0	144.3	1.23	1.29	+27.3 (26.9)	11.8	+15.5	1.96	389	499	10
Sample covariance, 120 months	191.7	148.8	1.28	1.28	+20.5 (27.9)	12.1	+8.4	2.02	455	537	7

Realised tracking error is the standard deviation of the monthly active return times the square root of 12 and the forecast the mean of the optimiser's own monthly forecasts; realised over budget divides the realised figure by the budget; the bias statistic is the standard deviation of the monthly active return divided by the same month's forecast (the forecast tracking error over the square root of twelve), 1 for a calibrated forecast. The standard error of an active return is its realised tracking error over the square root of 47 years. Trading cost is charged at 50 basis points per unit of one-way turnover and the net return is the gross less the mean cost, so it carries the gross return's standard error; the benchmark itself turns over 0.39 per cent a month. The budget binds in a month when the forecast tracking error is within one basis point of it; it is not met in a month in which the mandate's tracking-error-minimising portfolio already exceeds it, or the turnover cap stops the path short of it.

Table 2. What the budget bought: mean active exposures to the targeted factors, in beta units, and the exposure bought per basis point of realised tracking error

Covariance estimate	*Value (HML)*	*Profitability (RMW)*	*Investment (CMA)*	*Momentum (Mom)*	*The four together*	*Bought per bp, the four*	*Of which value*	*Profitability*	*Investment*	*Momentum*
Budget of 75 basis points a year										
Sample covariance, 3 years daily	+0.006	+0.025	+0.022	+0.004	+0.057	0.00057	0.00006	0.00025	0.00022	0.00004
Ledoit-Wolf shrinkage, 3 years daily	+0.003	+0.021	+0.020	+0.003	+0.047	0.00048	0.00003	0.00021	0.00021	0.00003
Six-factor model, 3 years daily	+0.010	+0.028	+0.022	+0.004	+0.064	0.00058	0.00009	0.00025	0.00020	0.00004
Principal components, 3 years daily	+0.009	+0.028	+0.022	+0.005	+0.064	0.00057	0.00008	0.00025	0.00020	0.00004
Sample covariance, 120 months	+0.005	+0.035	+0.041	-0.001	+0.081	0.00063	0.00004	0.00027	0.00032	0.00000
Budget of 100 basis points a year										
Sample covariance, 3 years daily	+0.021	+0.048	+0.039	+0.008	+0.116	0.00096	0.00017	0.00040	0.00033	0.00006
Ledoit-Wolf shrinkage, 3 years daily	+0.019	+0.045	+0.038	+0.007	+0.109	0.00092	0.00016	0.00038	0.00032	0.00006
Six-factor model, 3 years daily	+0.023	+0.051	+0.040	+0.009	+0.123	0.00092	0.00017	0.00038	0.00030	0.00007
Principal components, 3 years daily	+0.024	+0.053	+0.043	+0.009	+0.128	0.00094	0.00018	0.00038	0.00031	0.00006
Sample covariance, 120 months	+0.018	+0.056	+0.063	+0.002	+0.140	0.00093	0.00012	0.00038	0.00042	0.00001
Robust (worst case of the five)	+0.014	+0.040	+0.038	+0.004	+0.095	0.00082	0.00012	0.00034	0.00032	0.00004
RiskMetrics (exponentially weighted), 5 years daily	+0.022	+0.047	+0.041	+0.007	+0.116	0.00098	0.00018	0.00039	0.00034	0.00006
Sample covariance, 3 years daily, value alone	+0.074	-0.009	-0.037	-0.005	+0.023	0.00019	0.00060	-0.00007	-0.00030	-0.00004
Sample covariance, 3 years daily, momentum alone	-0.011	-0.007	-0.019	+0.039	+0.002	0.00002	-0.00009	-0.00006	-0.00015	0.00032
Budget of 150 basis points a year										
Sample covariance, 3 years daily	+0.035	+0.076	+0.064	+0.011	+0.186	0.00107	0.00020	0.00044	0.00037	0.00006
Ledoit-Wolf shrinkage, 3 years daily	+0.035	+0.075	+0.063	+0.011	+0.184	0.00107	0.00020	0.00044	0.00037	0.00006
Six-factor model, 3 years daily	+0.037	+0.079	+0.064	+0.011	+0.191	0.00102	0.00020	0.00042	0.00034	0.00006
Principal components, 3 years daily	+0.037	+0.078	+0.063	+0.012	+0.191	0.00103	0.00020	0.00042	0.00034	0.00007
Sample covariance, 120 months	+0.035	+0.082	+0.081	+0.004	+0.203	0.00106	0.00018	0.00043	0.00042	0.00002

An active exposure is the sum over industries of the active weight times the industry's beta to the factor, the betas from a regression of each industry's daily excess returns on the six factors over the three years before the month; its mean over the 566 months is shown, each targeted factor counted in the direction of its premium, and the four together are their sum, the mean factor objective on the seventeen paths that target all four; for the two single-factor paths the mean factor objective is the targeted factor's own column. The exposure bought per basis point divides the mean exposure by the path's realised tracking error in basis points a year.

3.2 What the budget cost

The realised tracking error exceeds the budget on every path. At 100 basis points the four daily estimators realise 118.6 to 137.0 basis points a year and the monthly covariance 150.1, ratios to the budget of 1.19 to 1.37 and 1.50; at 150 the ratios are 1.14 to 1.25 and 1.28, and at 75 they are 1.31 to 1.50 and 1.71. E3 expected 1.2 to 1.6 on the daily estimators and more on the monthly covariance at every budget: 9 of the 12 daily cells are inside the range and the three outside

are below it, 1.19 for shrinkage at 100 and 1.15 and 1.14 for the sample covariance and shrinkage at 150, while the monthly covariance's ratio is above the largest daily ratio at each budget, so E3 is not met, on the side of a smaller miss. The first note's portfolio is the in-sample minimum of a forecast, and a minimiser picks the weights for which the forecast is lowest relative to the truth (Michaud, 1989), which gave bias statistics of 1.22 to 1.39 on daily data; a portfolio held at a level by the budget is less of an in-sample minimum, so the forecast at the level misses by less, and the bias statistics of Table 1 are 1.16 to 1.35 on the daily estimators at 100. The monthly covariance is the worst calibrated at 75 and 100 basis points, as in the first note (1.45 at 100 against 1.61 under the mandate and 1.16 to 1.35 on the daily estimators here), and at 150 its 1.28 sits below the factor model's 1.31 and principal components' 1.29; the robust portfolio realises 115.6 basis points at a bias statistic of 1.08, again the best calibrated of the seven at 100.

The spread across the four daily estimators is 18.4 basis points at 100 on a level of about 130, against 8.8 on a level of about 90 under the first note's mandate, within the 20 that E7 allowed, and the ranking is the first note's: the sample covariance and shrinkage at the low end, the factor model and principal components at the high end, the monthly covariance behind them. With the exposure bought per basis point within 4.5 per cent of its mean across the four and the monthly covariance the worst calibrated of the five at 100, E7 is met: the estimator matters in what the budget cost in realised tracking error and within 4.5 per cent in what it bought. The turnover cap binds in 378 to 518 of the 566 months at 100 basis points on the five estimators, against 1.78 to 2.00 per cent of one-way turnover a month and the benchmark's own 0.39, and at 50 basis points per unit of turnover the cost is 10.7 to 12.0 basis points a year.

3.3 What the budget earned

Table 3 reports the attribution. The gross active return at 100 basis points is +8.9, +6.5, +15.5, +9.9 and +6.0 basis points a year under the sample covariance, shrinkage, the factor model, principal components and the monthly covariance, on standard errors of 17.6, 17.3, 19.5, 19.9 and 21.9; at 150 it is +20.5 to +34.0 on standard errors of 25 to 28, and at 75 between minus 3.9 and +3.7 on 14 to 19. The largest ratio of an active return to its standard error across the nineteen paths is 1.36, shrinkage at 150, so no path's active return is beyond two standard errors of zero. E5's sign clause is met, the gross return positive on 5 of 5 at 100, and its cost clause is not: the cost of turnover at 50 basis points per unit is 10.7 to 12.0 basis points a year, within 10 of the gross on 0 of 5, so the net active return is minus 6.0 to +4.8. The sign of the gross return follows from the targets and the factors' realised returns over this sample; the study estimates no premium and the sign says nothing about another sample.

E4 is met in its first clause and not in its second, and the way the second fails is the attribution's main finding about what a factor objective buys on industries. The four targeted contributions sum to +26.7 to +35.5 basis points a year at 100 on the five estimators and +49.9 to +58.9 at 150, on standard errors of 11 to 21, beyond two standard errors on all ten of those paths; at 75 they sum to +5.7 to +16.6 on standard errors of 8 to 13, inside two on all five; positive on 15 of 15 main paths. The factor share is 0.14 to 0.28 on the ten paths at 100 and 150, where the expectation was at least 0.30: the range of the first note's incidental exposures, 0.14 to 0.25 under the mandate, unchanged by exposures three to seven times their size. Table 4 says why. The six contributions' own variance is 34 to 52 per cent of the active return's, above the level on every path; the residual's is 72 to 86 per cent; and the two move against each other, their covariance taking 18 to 26 per cent off the sum, so that one minus the residual's part is 14 to 28 per cent. The industries' own returns are the larger part of a factor objective's active return on 49 industries, and when the factors pay, the industries that carry the exposures give some of it back: the factor part earned +37 to +47 basis points a year at 100 and the residual took back 31 to 37 of it, and the residual is negative on every one of the nineteen paths, minus 10 to minus 47. The first note found the mirror image under the mandate, a factor part of minus 11 to minus 17 and a residual of +7 to +14 on an active return near zero; the objective turned both parts round and left the sum inside its standard error. The level of 0.30 was written from the first note's shares and the sentence that deliberate exposures raise the share; they raised the factor part's variance and the residual's with it, because the same active weights carry both.

The two factors outside the objective are not at zero either. The active exposure to the market factor is 0.012 to 0.015 at 100 basis points on the four daily estimators and 0.003 on the monthly covariance, where the first note's mandate held it within 0.0015 of zero on the daily estimators, and it contributed +9.8 to +11.7 basis points a year on standard errors of 3.5 to 4.2, the monthly covariance minus 0.5: beta neutrality is to the benchmark, the cap-weighted 49, and the industries the objective buys move more with French's market factor than the ones it sells. Size contributed +1.2 to +2.8 at 100. The targeted total excludes both.

Table 3. Where the active return came from: the six factor contributions, the targeted total, the residual and the active return, in basis points a year

Covariance estimate	*Mkt-RF*	*SMB*	*HML*	*RMW*	*CMA*	*Mom*	*Targeted total (s.e.)*	*Factor total (s.e.)*	*Residual (s.e.)*	*Active return (s.e.)*
Budget of 75 basis points a year										
Sample covariance, 3 years daily	+5.5	+1.4	-4.0	+3.2	+8.4	+0.9	+8.5 (8.4)	+15.5 (8.3)	-17.7 (13.1)	-2.2 (14.6)
Ledoit-Wolf shrinkage, 3 years daily	+5.1	+2.0	-3.5	+1.5	+7.8	-0.1	+5.7 (8.3)	+12.8 (8.2)	-15.8 (12.8)	-3.1 (14.4)
Six-factor model, 3 years daily	+7.0	+1.5	-4.4	+5.1	+9.4	+1.5	+11.6 (8.8)	+20.1 (8.6)	-16.4 (14.9)	+3.7 (16.1)
Principal components, 3 years daily	+6.0	+0.4	-5.3	+4.2	+11.4	+1.1	+11.4 (9.6)	+17.8 (9.4)	-21.6 (14.7)	-3.9 (16.4)
Sample covariance, 120 months	-2.3	+2.2	+2.6	+6.3	+13.6	-5.9	+16.6 (13.2)	+16.4 (12.2)	-17.3 (16.6)	-0.9 (18.6)
Budget of 100 basis points a year										
Sample covariance, 3 years daily	+10.2	+2.3	-0.9	+13.5	+12.6	+3.9	+29.1 (11.1)	+41.6 (10.8)	-32.7 (15.9)	+8.9 (17.6)
Ledoit-Wolf shrinkage, 3 years daily	+9.8	+2.8	-1.0	+12.3	+12.2	+3.2	+26.7 (10.9)	+39.2 (10.7)	-32.8 (15.6)	+6.5 (17.3)
Six-factor model, 3 years daily	+11.7	+2.3	-1.5	+16.0	+13.6	+5.2	+33.3 (11.8)	+47.3 (11.4)	-31.8 (18.0)	+15.5 (19.5)
Principal components, 3 years daily	+10.6	+1.2	-2.9	+16.9	+16.6	+4.9	+35.5 (12.4)	+47.3 (12.1)	-37.4 (18.0)	+9.9 (19.9)
Sample covariance, 120 months	-0.5	+2.3	+6.2	+15.6	+18.5	-5.4	+34.8 (16.7)	+36.7 (15.3)	-30.7 (19.1)	+6.0 (21.9)
Robust (worst case of the five)	+8.5	+0.5	+1.0	+8.8	+12.1	-0.2	+21.7 (11.1)	+30.7 (10.7)	-28.4 (15.1)	+2.3 (16.8)
RiskMetrics (exponentially weighted), 5 years daily	+8.7	+1.1	-0.2	+14.1	+11.3	+3.0	+28.2 (10.8)	+38.1 (10.5)	-30.6 (15.9)	+7.5 (17.3)
Sample covariance, 3 years daily, value alone	+9.3	+1.1	+8.7	-5.1	-3.9	-8.0	-8.3 (11.3)	+2.0 (12.3)	-16.5 (15.2)	-14.5 (17.9)
Sample covariance, 3 years daily, momentum alone	-4.8	+3.4	-11.7	-3.8	+2.8	+32.2	+19.5 (10.1)	+18.1 (10.7)	-10.3 (16.0)	+7.8 (17.9)
Budget of 150 basis points a year										
Sample covariance, 3 years daily	+13.8	+8.1	+2.7	+23.9	+20.0	+4.2	+50.8 (15.9)	+72.7 (16.1)	-38.9 (22.7)	+33.9 (25.2)
Ledoit-Wolf shrinkage, 3 years daily	+14.3	+8.3	+2.7	+23.5	+19.6	+4.1	+49.9 (15.8)	+72.5 (15.9)	-38.5 (22.5)	+34.0 (24.9)
Six-factor model, 3 years daily	+15.9	+8.7	+2.2	+26.3	+22.0	+5.0	+55.5 (16.4)	+80.1 (16.7)	-46.3 (24.5)	+33.9 (27.3)
Principal components, 3 years daily	+14.8	+6.7	-0.8	+25.3	+23.0	+5.2	+52.8 (16.6)	+74.3 (16.9)	-47.0 (24.2)	+27.3 (26.9)
Sample covariance, 120 months	-3.3	+7.5	+13.6	+26.1	+23.4	-4.2	+58.9 (21.5)	+63.1 (20.2)	-42.6 (23.8)	+20.5 (27.9)

A contribution is the active exposure to the factor, with the betas estimated before the month, times the factor's return in the month, averaged and stated per year; the targeted total is the sum of the four targeted factors' contributions, the factor total the sum of all six, and the residual is the active return minus the six contributions, the part the industries' own returns account for. The identity, contributions plus residual equal the active return, holds to 1.7e-18 in every month of every path. Standard errors are the standard deviation of the monthly series over the square root of the number of months, times 12. Mkt-RF is the market factor, SMB size, HML value, RMW profitability, CMA investment and Mom momentum (Fama and French, 2015; Carhart, 1997).

3.4 The gap, the single-factor paths and the decades

Table 4 reports the gap. The gap per unit of exposure bought is 0.039 to 0.044 at 100 basis points on the five estimators, 0.034 to 0.037 at 150 and 0.054 to 0.076 at 75: the betas estimated after the month take back about 4 per cent of the exposure chosen before it at the main budget, and a larger fraction at the small budget, where the exposure bought is small and the gap, which comes from the betas' movement and not from the budget, is not. The largest single-month gaps fall in 1990-10 on eleven of the nineteen paths, in 2005-07 on five and in 1981-09 and 1987-10 on the others, months in which the three-year beta windows turned over a shock. E6 expected momentum alone to have the higher one-way turnover, the larger gap and the more often binding cap than value alone: its turnover is 1.94 per cent a month against 1.87 and its cap binds in 478 months against 408, as expected, and its gap is 0.00185 in beta units against value alone's 0.00235, the other way, so E6 is not met in its gap clause. Per unit of what was bought the order is the expected one, 0.047 against 0.032, because momentum alone bought half the exposure; the sentence that momentum betas rotate faster holds per unit and not in beta units.

By decade, at 100 basis points, the targeted total was negative in 1979 to 1989 on four of the five estimators, minus 6 to minus 18 basis points a year, and largest in the 1990s, +70 to +93, when the residual was at its most negative, minus 104 to minus 143, so that the active return of that decade is between minus 17 and +14: the decade in which the factors paid most is the decade in which the industries carrying the exposures gave most of it back. In the 2010s the factor part lies within 7 basis points of zero on the five estimators and the residual between minus 3 and minus 17. The by-decade table, `factor_objective_attribution_by_decade_202608.csv`, describes where the contributions came from and carries no claim about any decade.

Table 4. The factor share and the three shares of the active return's variance behind it, and the gap per unit of exposure bought

Covariance estimate	*Factor share*	*Factor total's own variance*	*Residual's own variance*	*Twice their covariance*	*Mean absolute gap, beta units*	*Mean factor objective*	*Gap per unit*	*Largest gap, month*
Budget of 75 basis points a year								
Sample covariance, 3 years daily	0.20	0.33	0.80	-0.13	0.00371	0.0571	0.065	1981-09
Ledoit-Wolf shrinkage, 3 years daily	0.20	0.33	0.80	-0.13	0.00359	0.0471	0.076	1981-09
Six-factor model, 3 years daily	0.14	0.29	0.86	-0.15	0.00427	0.0641	0.067	1990-10
Principal components, 3 years daily	0.20	0.33	0.80	-0.13	0.00411	0.0642	0.064	1990-10
Sample covariance, 120 months	0.21	0.43	0.79	-0.22	0.00441	0.0810	0.054	2005-07
Budget of 100 basis points a year								
Sample covariance, 3 years daily	0.18	0.38	0.82	-0.20	0.00468	0.1158	0.040	1990-10
Ledoit-Wolf shrinkage, 3 years daily	0.18	0.38	0.82	-0.20	0.00455	0.1089	0.042	2005-07
Six-factor model, 3 years daily	0.14	0.34	0.86	-0.20	0.00534	0.1227	0.044	1990-10
Principal components, 3 years daily	0.19	0.37	0.81	-0.18	0.00521	0.1283	0.041	1990-10
Sample covariance, 120 months	0.23	0.49	0.77	-0.26	0.00544	0.1401	0.039	2005-07
Robust (worst case of the five)	0.19	0.41	0.81	-0.21	0.00448	0.0953	0.047	2005-07
RiskMetrics (exponentially weighted), 5 years daily	0.15	0.37	0.85	-0.22	0.00480	0.1162	0.041	2005-07
Sample covariance, 3 years daily, value alone	0.27	0.47	0.73	-0.20	0.00235	0.0741	0.032	1990-10
Sample covariance, 3 years daily, momentum alone	0.20	0.36	0.80	-0.16	0.00185	0.0391	0.047	1987-10
Budget of 150 basis points a year								
Sample covariance, 3 years daily	0.19	0.41	0.81	-0.22	0.00645	0.1856	0.035	1990-10
Ledoit-Wolf shrinkage, 3 years daily	0.19	0.41	0.81	-0.22	0.00636	0.1838	0.035	1990-10
Six-factor model, 3 years daily	0.20	0.37	0.80	-0.18	0.00710	0.1906	0.037	1990-10
Principal components, 3 years daily	0.19	0.40	0.81	-0.20	0.00684	0.1908	0.036	1990-10
Sample covariance, 120 months	0.28	0.52	0.72	-0.25	0.00694	0.2026	0.034	1990-10

The factor share is one minus the variance of the residual over the variance of the active return. The three shares are the factor total's own variance, the residual's own variance and twice their covariance, each over the active return's variance; they sum to one. The gap of a month is the factor objective chosen with the betas estimated before the month minus the value the same active weights carry with the betas estimated after it; the mean absolute gap is over the 565 months that have a following month, and the gap per unit divides it by the path's mean factor objective.

Table 5. The seven expectations, written before any path ran, and what the run found

Expectation	*Written before the run*	*Found, in the words of the notebook that judged it*	*Verdict*
E1	At 100 basis points the budget binds, within one basis point, in at least 95 per cent of the months the mandate alone can meet, for each of the four daily estimators.	binding share over the months the mandate alone can meet, four daily estimators at 100 bp: 0.9698, 0.9732, 0.9658, 0.9688	MET
E2	At 100 basis points every targeted exposure is positive on the five estimators, the average of the four is at least 0.024 for the sample covariance, and the average rises with the budget on every estimator.	every targeted exposure positive on 5 of 5. average of the four for the sample covariance 0.0289. rises with the budget for 5 of 5	MET
E3	Realised tracking error exceeds the budget by 20 to 60 per cent on the daily estimators, and by more on the monthly covariance, at every budget.	daily ratios inside the range in 9 of 12. monthly above the largest daily at 3 of 3 budgets	NOT MET
E4	The four targeted contributions sum to a positive amount a year on every main path, and at 100 and 150 basis points the six factors explain at least 30 per cent of the active return's variance on every main path.	targeted total positive on 15 of 15 main paths. factor share at least 0.30 on 0 of 10 main paths at 100 and 150 bp (0.1439 to 0.2751)	NOT MET
E5	At 100 basis points the gross active return is positive on at least four of the five estimators, and the net return at 50 basis points per unit of turnover lies within 10 basis points a year of the gross.	gross positive for 5 of 5. net within 10 bp for 0 of 5	NOT MET
E6	Momentum alone has a higher one-way turnover and a larger gap than value alone, and the turnover cap binds in more months on the momentum path.	turnover higher, cap binds more, gap not larger on the momentum path	NOT MET
E7	At 100 basis points the realised tracking errors of the four daily estimators lie within 20 basis points of each other, the exposure bought per basis point differs across them by less than 30 per cent of its mean, and the monthly covariance is the worst calibrated of the five.	spread 18.41 bp. exposure per bp differs by 0.0447 of the mean. worst calibrated: the sample covariance, 120 months (1.45)	MET

E1 to E7 as written in `src/bp/constants.py` on 8 October 2026; E1 and E2 were restated on 9 October from the arithmetic of notebook 14 at the three report months, before any path ran, and the levels of 8 October stay in the file marked as superseded. E1, E2, E3, E5, E6 and E7 are judged in notebook 15 and E4 in notebook 16; the verdict files are `factor_objective_verdicts_202608.csv` and `factor_objective_attribution_verdict_202608.csv`.

4. Discussion

The natural comparison is with the first note's mandate on the same months. There the tracking error was the cost of rules nobody chose for return, 84 to 96 basis points a year, and the factor exposures it left behind cost 9 to 13 basis points a year under the tilt alone while the industries' own returns paid 12 to 20. Here the tracking error is spent on purpose, and the estimates indicate that the purchase works as arithmetic says it should and earns what the sample's factor returns imply: a budget of 100 basis points buys 0.109 to 0.140 of exposure in beta units, the four targeted factors contribute +26.7 to +35.5 basis points a year, and the exposures to the two factors outside the objective, the market factor above all, contribute another +12 to +14 on the four daily estimators and +2 on the monthly covariance. The same comparison shows what the purchase does not change. The industries' own returns remain the larger part of the active return's variance, 72 to 86 per cent, and they move against the factor part, so the factor share stays where the mandate left it, and the active return itself, +6 to +16 basis points a year gross at 100 on standard errors of 17 to 22, is inside two standard errors of zero on every path, as the mandate's was inside one. The estimator's role is the first note's as well: the four daily estimators differ by 18.4 basis points in what the budget cost in realised tracking error and by 4.5 per cent in what it bought, the monthly covariance is the worst calibrated at the main budget and the robust portfolio the best.

For the holder of such a mandate the estimates therefore suggest three things. A budget of forecast tracking error spent on the factor objective buys most of its exposure in the two factors whose industry betas the optimiser can raise cheaply within the bounds, profitability and investment, and little momentum, so an equal weighting in beta units should be read as a statement about the objective and not about the portfolio. The realised tracking error of a portfolio held at a budget by the optimiser's own forecast exceeds the budget by 14 to 50 per cent on the daily estimators and 28 to 71 per cent on the monthly covariance, in the same direction as the mandate's minimiser missed, and by less at 100 basis points by the bias statistic (1.16 to 1.35 against 1.22 to 1.39), so a budget written on the forecast is a budget on something smaller than what is realised. And the return the exposures earn over 47 years is of the size and sign the sample's factor returns imply, beyond two standard errors for the targeted part at 100 and 150 basis points, and the active return a client is paid is that part less what the industries carrying the exposures gave back, inside two standard errors of zero on every path; the attribution indicates where the difference went and estimates no premium.

At the same time, the design has limits, each with its cause. First, the assets are industry portfolios, so a basis point of realised tracking error buys about 0.001 of exposure in beta units, 0.03 per factor at 100 basis points, and the mandate's bounds stop the move at 150 in 104 to 201 months; whether a stock-level mandate, with hundreds of positions and a bound a fraction of this one, buys more exposure per basis point or a different split is a question these data cannot answer. Second, the betas are estimated on three years of daily returns, fixed before the code, so the exposure bought and the gap both depend on that window; a longer window would change both, and by how much is not measured here, and the attribution on the betas estimated after the month would move each contribution by the gap of that month, which Table 4 states on average. Third, the factor share has more than one definition: defined as the six contributions' own variance over the active return's it would be 0.34 to 0.52 and above the level, and defined as the first note defined it, one minus the residual's part, it is 0.14 to 0.28; the definition was written first and the verdict follows it, and the decomposition is reported so that a reader can judge either. Fourth, whether the targeted total's sign would hold over another sample is not a question these 566 months answer, since the sum is the realised factor returns times the exposures and the study estimates no premium. Finally, French revises history, so a confirmation run on a later vintage will reproduce every verdict line and every printed decimal and not every digit, and the provenance record says which vintage a run used.

What would settle the open questions is a stock-level panel with the same mandate and budget, which would require a point-in-time constituent list and market capitalisations, and an attribution on betas estimated with a longer window beside the three-year one, which would require the gap to be measured under both. Against this evidence, the claim is a narrow one: on French's 49 industries, under a mandate that holds the portfolio within 2 percentage points of its benchmark, trades no more than 2 per cent a month and reduces three industries by half, a budget of forecast tracking error spent on the factor objective bought the four exposures it targeted, most of them in profitability and investment, at a realised tracking error 14 to 71 per cent above the budget; the targeted factors' contributions were positive on every main path and beyond two standard errors at 100 and 150 basis points, the industries' own returns took back most of the factor part, the six factors explain 14 to 28 per cent of the active return's variance, and no path's active return is beyond two standard errors of zero.

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